

Reframing preprocessing selection as model-internal calibration in near-infrared spectroscopy

A large-scale benchmark of operator-adaptive PLS and Ridge models

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Abstract

Near-infrared spectroscopy (NIRS) is a rapid and non-destructive analytical technique, but the development of reliable calibration models remains strongly dependent on spectral preprocessing. In routine workflows, preprocessing is commonly selected through large external pipeline searches, which are computationally costly, statistically unstable on small calibration sets, and difficult to audit. Here, we introduce operator-adaptive calibration, a framework that embeds linear preprocessing selection inside the calibration model itself. Candidate spectral treatments are represented as linear operators acting on the spectra, while nonlinear or sample-adaptive corrections such as SNV, MSC and ASLS are handled as fold-local branches to avoid leakage.

The framework is instantiated for PLS and Ridge regression. For PLS, operator selection is implemented through covariance identities that enable fast NIPALS and SIMPLS variants while preserving original-wavelength coefficients. For Ridge, operator-adaptive kernels provide a dual formulation with recoverable original-space coefficients. The approach was evaluated on a heterogeneous benchmark of more than 50 NIRS datasets and compared with conventional PLS, Ridge, CatBoost and CNN baselines under documented search budgets.

Compact operator-adaptive PLS combined with ASLS branch preprocessing achieved a multi-seed median RMSEP/PLS ratio of 0.962 with 37 wins on 53 paired datasets. In the current partial AOM-Ridge overlap, the Blender selector achieved a median RMSEP/Ridge-HPO ratio of 0.905 with 4 wins on 4 paired datasets; the full-cohort Blender/AutoSelect promotion remains pending. Beyond predictive accuracy, the proposed models reduce the dependence on large preprocessing-HPO campaigns, produce traceable operator choices, retain interpretable coefficients, and achieve seconds-scale fitting for compact AOM-PLS. Operator-adaptive calibration therefore provides a practical route toward faster, more robust and auditable NIRS method development.

Keywords: near-infrared spectroscopy; analytical calibration; chemometrics; preprocessing selection; partial least squares; ridge regression; model robustness; spectral data analysis; interpretable machine learning.

1 Introduction

Near-infrared spectroscopy is widely used as a rapid, non-destructive analytical technique for food, agriculture, plant phenotyping and process control [1, 2]. Yet the development of reliable

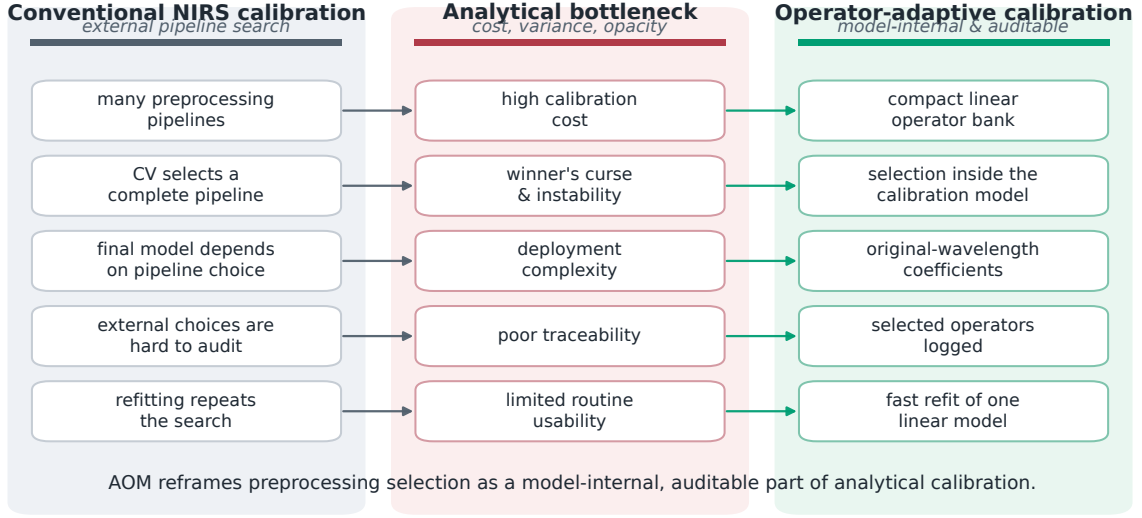


Figure 1: From external preprocessing search to operator-adaptive analytical calibration. The proposed workflow converts preprocessing alternatives from competing full pipelines into structured model components whose selections can be logged, audited and refitted.

calibration models remains a major bottleneck. In routine practice, predictive performance often depends less on the regression algorithm itself than on the selection of spectral preprocessing. This selection is usually performed as an external pipeline search over smoothing, derivatives, scatter correction, baseline correction and scaling choices. As a result, calibration development can become computationally expensive, statistically unstable on small calibration sets, and difficult to audit.

We argue that preprocessing selection should be treated as a core part of analytical calibration rather than as an external hyperparameter search. This is particularly important for NIRS, where calibration models are frequently deployed across heterogeneous matrices, traits, instruments and sample sizes. Preprocessing selection conditions not only predictive performance, but also external robustness, reproducibility, development cost, transferability to non-expert users and the auditability of the final calibration object.

The novelty of the present work is therefore not another preprocessing filter. It is a change in the calibration paradigm: preprocessing alternatives are treated as structured model components with recoverable coefficients, rather than as competing black-box pipelines. We call this approach operator-adaptive calibration. Strict linear spectral treatments are encoded as a bank of operators $\{\mathbf{A}_1, \dots, \mathbf{A}_m\}$ acting on the spectra, and the calibration model selects among them internally. The identity operator is always included, so standard PLS or Ridge remains available. Nonlinear or sample-adaptive corrections such as SNV, MSC and ASLS are kept as explicit fold-local branches to avoid leakage and to preserve the distinction between linear operator algebra and fitted preprocessing.

This study makes four contributions to analytical NIRS calibration:

1. It introduces operator-adaptive calibration, a framework in which linear spectral preprocessing candidates are embedded as model components rather than selected as external pipelines.
2. It provides fast PLS and Ridge instantiations that retain original-wavelength coefficients and therefore remain interpretable and deployable.

3. It validates the approach on a heterogeneous benchmark of more than 50 NIRS datasets, covering diverse sample sizes, response scales and spectral structures.
4. It shows that operator-adaptive calibration can improve predictive performance while reducing calibration search complexity and increasing traceability.

Unlike many NIRS calibration studies that focus on one matrix or one analytical trait, we evaluate the proposed strategy across a heterogeneous multi-dataset benchmark. This design tests whether the method improves the calibration process itself, rather than optimizing a single application-specific case. Our central claim is that operator-adaptive calibration provides a practical alternative to large preprocessing-HPO campaigns for NIRS: it improves or maintains predictive performance across heterogeneous datasets, reduces calibration search complexity, preserves linear interpretability, and produces traceable preprocessing decisions suitable for analytical deployment.

2 Analytical calibration problem and rationale

2.1 PLS and Ridge as linear calibration references

PLS regression constructs latent variables that maximize covariance between spectra and responses, and has become a standard tool in chemometrics [3, 4]. NIPALS provides an iterative extraction view, while SIMPLS operates directly on covariance structures [5]. For a centered matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ and response $\mathbf{Y} \in \mathbb{R}^{n \times q}$, a PLS model with weights $\mathbf{W}$, loadings $\mathbf{P}$ and response loadings $\mathbf{Q}$ has regression coefficients

$$\mathbf{B} = \mathbf{W}(\mathbf{P}^T \mathbf{W})^+ \mathbf{Q}^T, \tag{1}$$

where $+$ denotes the Moore–Penrose inverse.

Ridge regression estimates a linear coefficient matrix under an L_2 penalty [6]. It is less tied to chemometric tradition than PLS, but is a strong baseline for high-dimensional collinear predictors because shrinkage stabilizes the inverse problem. In dual form, ridge depends on the sample Gram matrix $\mathbf{X}_c \mathbf{X}_c^T$ and is naturally connected to kernel methods [7, 8].

2.2 Preprocessing as model selection

Spectral preprocessing aims to reduce nuisance variation or enhance relevant chemical contrast. Savitzky–Golay filters smooth spectra and estimate derivatives [9]; Norris–Williams derivatives are a related finite-window approach [10]. SNV and MSC reduce scatter effects [11, 12], while EMSC extends this idea with additional terms [13]. Baseline methods such as asymmetric penalized least squares correct broad low-frequency drifts [14, 15].

These procedures are scientifically meaningful, but their combinations create a model-selection problem. A recent local NIRS benchmark protocol used hierarchical preprocessing search spaces in which PLS and Ridge each evaluated approximately 5400 candidate fold-level configurations per dataset, CNN-1D approximately 2700, and CatBoost approximately 90. Such budgets are defensible for a benchmark, but they are costly for routine calibration and increase the risk that the selected pipeline reflects validation noise.

2.3 Modern baselines

Gradient boosting methods such as CatBoost [16] and one-dimensional convolutional neural networks [17, 18] provide useful nonlinear baselines for NIRS. They are valuable comparison points because they represent two common directions away from classical chemometrics: tabular machine learning and spectral deep learning. In this paper they are used as baselines, while

the methodological focus remains on linear, coefficient-bearing calibrations. Previous work by the authors has also investigated deep learning for NIRS phenotyping and food-quality traits [19–21].

3 Operator-adaptive calibration framework

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ denote spectra and $\mathbf{Y} \in \mathbb{R}^{n \times q}$ denote one or several responses. An operator bank is a finite set

$$\mathcal{A} = \{\mathbf{A}_1, \dots, \mathbf{A}_m\}, \quad \mathbf{A}_b \in \mathbb{R}^{p \times p}. \quad (2)$$

Each operator acts on spectra by right multiplication:

$$\mathbf{X}_b = \mathbf{X} \mathbf{A}_b^\top. \quad (3)$$

The bank contains the identity, ensuring that the unmodified calibration model is included. Typical strict linear operators include smoothing matrices, finite differences, Savitzky–Golay derivatives and detrending projections.

Several selection regimes are possible. In global selection, one operator is chosen for the complete model. In component-wise selection, a different operator may be selected for each latent component; we treat this per-operator-per-component variant as exploratory in the present paper because it can increase selection variance. In a superblock formulation, all operator-transformed views are concatenated or combined with weights. Finally, nonlinear or sample-adaptive transformations can be placed outside the strict operator bank as fold-local branches.

The strict linear distinction matters. Some preprocessing procedures are linear only after quantities have been estimated from the calibration fold; others are nonlinear in each sample. SNV, for example, subtracts and divides by sample-specific statistics. MSC and EMSC estimate correction parameters relative to a reference. ASLS estimates a baseline through an asymmetric iterative procedure. These operations remain useful in chemometric workflows, but treating them as fixed $\mathbf{A}_b$ matrices would obscure their statistical role and risk leakage if fitted outside the fold.

4 Model instantiations

4.1 Operator-adaptive PLS

4.1.1 Standard PLS notation

For centered $\mathbf{X}$ and $\mathbf{Y}$, PLS extracts latent scores $\mathbf{T} = \mathbf{X} \mathbf{W}^*$ and fits $\mathbf{Y}$ through these scores. In the notation used here, the coefficient matrix is

$$\mathbf{B} = \mathbf{W}(\mathbf{P}^\top \mathbf{W})^+ \mathbf{Q}^\top, \quad (4)$$

where $\mathbf{W}$ are weight vectors, $\mathbf{P}$ are X -loadings and $\mathbf{Q}$ are Y -loadings. Predictions for new spectra are then

$$\hat{\mathbf{Y}} = (\mathbf{X}_{\text{new}} - \bar{\mathbf{X}}) \mathbf{B} + \bar{\mathbf{Y}}. \quad (5)$$

4.1.2 Operator-adaptive PLS coefficients

For a selected operator $\mathbf{A}_b$, a PLS weight vector r_a estimated in the transformed space corresponds to an original-space effective direction

$$z_a = \mathbf{A}_b^\top r_a. \quad (6)$$

Collecting these directions in $\mathbf{Z} = [z_1, \dots, z_k]$ yields scores

$$\mathbf{T} = \mathbf{XZ}. \quad (7)$$

The original-space coefficient matrix is

$$\mathbf{B} = \mathbf{Z}(\mathbf{P}^\top \mathbf{Z})^+ \mathbf{Q}^\top. \quad (8)$$

This is the key interpretability property: even if operators are evaluated internally, the final model is a single linear map from original spectra to responses.

4.1.3 Fast covariance and adjoint implementations

The materialized reference constructs each $\mathbf{XA}_b^\top$ and fits ordinary PLS. This is useful for testing but wasteful when many operators are considered. For strict linear operators, candidate covariance matrices can be obtained without materializing transformed spectra:

$$(\mathbf{XA}_b^\top)^\top \mathbf{Y} = \mathbf{A}_b \mathbf{X}^\top \mathbf{Y}. \quad (9)$$

Equation 9 directly supports a covariance-space SIMPLS engine. A NIPALS implementation can similarly apply adjoint operators to covariance vectors and then map transformed weights back to original-space directions. The implementation used in the experiments includes tests that identity-only AOM equals standard PLS, fixed-operator materialized and fast forms agree numerically, and NumPy/PyTorch engines are consistent.

4.2 Operator-adaptive Ridge

Ridge regression provides a complementary view because it depends on spectral geometry rather than on the cross-covariance alone. Let $\mathbf{X}_c$ and $\mathbf{Y}_c$ be centered calibration matrices. For a single operator,

$$\mathbf{Z}_b = \mathbf{X}_c \mathbf{A}_b^\top, \quad (10)$$

and the dual ridge kernel is

$$\mathbf{K}_b = \mathbf{Z}_b \mathbf{Z}_b^\top = \mathbf{X}_c \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}_c^\top. \quad (11)$$

For regularization parameter $\alpha > 0$,

$$\mathbf{C} = (\mathbf{K}_b + \alpha \mathbf{I}_n)^{-1} \mathbf{Y}_c, \quad (12)$$

and the original-space coefficient matrix is

$$\beta_b = \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}_c^\top \mathbf{C}. \quad (13)$$

The superblock case combines several operators by concatenating transformed blocks with weights s_b :

$$\Phi(\mathbf{X}_c) = [s_1 \mathbf{X}_c \mathbf{A}_1^\top \mid \dots \mid s_m \mathbf{X}_c \mathbf{A}_m^\top]. \quad (14)$$

The resulting kernel is a sum of operator geometries,

$$\mathbf{K} = \sum_{b=1}^m s_b^2 \mathbf{X}_c \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}_c^\top, \quad (15)$$

and coefficients are recovered as

$$\beta = \mathbf{M} \mathbf{X}_c^\top \mathbf{C}, \quad \mathbf{M} = \sum_{b=1}^m s_b^2 \mathbf{A}_b^\top \mathbf{A}_b. \quad (16)$$

This highlights the conceptual difference between the two models. AOM-PLS uses the operator action on $\mathbf{X}^\top \mathbf{Y}$ to select covariance directions. AOM-Ridge uses the operator action on $\mathbf{X} \mathbf{X}^\top$ to select a regularized geometry. Both remain linear in the original spectra.

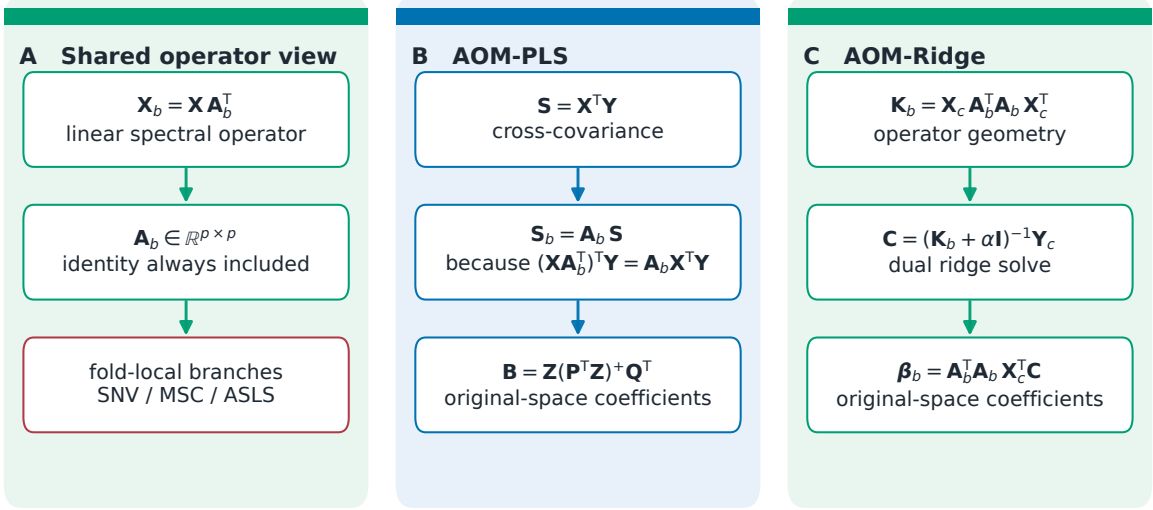


Figure 2: AOM mathematical structure. AOM-PLS exploits an operator identity on cross-covariances, while AOM-Ridge builds operator-specific sample kernels. Both recover original-space coefficients.

5 Implementation and software availability

AOM-PLS and POP-PLS are available in the open-source `nirs4all` Python library as regressor and classifier classes for global and component-wise operator selection. The same library provides spectral preprocessing, splitting utilities and pipeline serialization. The desktop application `nirs4all Studio` uses the same library stack and exposes these workflows through a graphical interface for users who do not write Python scripts. The research implementations and benchmark scripts used for the present AOM-PLS and AOM-Ridge analyses are kept in the public repository under `bench/AOM_v0`. A dedicated `bench/AOM_lib` implementation now provides the AOM-PLS compact bank through C++, Python, R, Julia, MATLAB and JavaScript/WASM bindings, with local C++ and Python parity validation against the research reference. Table 1 summarizes the current artifacts.

6 Benchmark design and validation protocol

6.1 Datasets and splits

We used the local multi-dataset NIRS benchmark assembled in the repository. The benchmark contains heterogeneous regression and classification datasets covering a wide range of sample sizes, wavelength counts and response scales. The present paper focuses on regression because AOM-Ridge is currently a regression method and because the strongest AOM-PLS evidence is regression based. Where an original train/test split existed, it was preserved. When no external split was provided, deterministic SPXY or stratified variants were used [22, 23]. The external test set was kept untouched during preprocessing and hyperparameter selection.

6.2 Model selection

For all compared models, preprocessing and hyperparameter selection were performed inside the calibration set. PLS selected the number of latent components by cross-validation. Ridge selected the regularization parameter α . CatBoost and CNN-1D followed the local reference

Table 1: Software availability. The production package contains the current AOM-PLS family; the AOM-Ridge experiments are distributed as research code in the benchmark directory pending API stabilization.

Component	Language	Lines	TestsStatus
nirs4all AOM-PLS (reference)	Python	1,600	tests/production/operators/models/test_aom_pls*
nirs4all AOM-lib wrapper	Python	50	tests/integration/operators/models/test_aom_pls*
AOM_lib C++17 core	C++17	1,400	cpp/build/test/operators, test_parity_kfold
AOM_lib C ABI	C/C++ extern "C"	250	cpp/build/test/smoke
AOM_lib Python binding	Python + pybind11	400	python/test/test_parity.py (9/9)
AOM_lib R binding	R + Rcpp	600	r/aompls/test/testthat/test_parity.R
AOM_lib MATLAB binding	MATLAB + MEX	400	matlab/test/parity/tested locally
AOM_lib Julia binding	Julia + ccall	300	julia/aompls/test/testthat/test.jl
AOM_lib JS/WASM binding	JavaScript + Emscripten	400	js/test/parity.mjs tested locally
AOM-Ridge research code	Python	8,800	bench/AOM_v0/Ridge/tests/

Status legend: *production* – shipping in **nirs4all** releases; *integration* – new wrapper exposing the C++ core through the Python pipeline API; *parity-verified* – numerical agreement with the Python AOM_v0 reference on BEER/CORN/ALPINE $\times$ kfold5/kfold5+oneSE/spxy5 (coefficient tolerance $< 10^{-8}$, prediction tolerance $< 10^{-8}$, RMSE-curve tolerance $< 10^{-9}$); *smoke-tested* – builds and runs a minimal end-to-end fit/predict against the C ABI; *code-ready, not tested locally* – bindings compile and ship with parity tests, but the host interpreter/toolchain (R, MATLAB, Julia, Emscripten) was not available in the current validation environment; *research* – experimental code distributed under **bench/AOM_v0** pending API stabilization. All artifacts live in the public repository under **bench/AOM_lib** and **bench/AOM_v0**.

benchmark protocol, with CatBoost using fixed hyperparameters under preprocessing selection and CNN-1D using a predefined architecture search space. AOM-PLS selected among operators in the compact bank and the number of components. AOM-Ridge selected operator geometries and regularization in its nested protocol.

The compact AOM operator bank is shown in Table 3. It is intentionally small: previous internal experiments found that larger banks can degrade external-test performance despite improving validation scores, which is consistent with selection variance and winner’s-curse effects in small- n settings.

6.3 Metrics

Regression performance is reported primarily through RMSEP on the independent test set. Because absolute errors are not comparable across chemical traits and response scales, dataset-level ratios or relative deltas are used for aggregate comparisons. For example, RMSEP/PLS below one indicates improvement over the PLS reference on the same dataset. We also report win counts, i.e., the number of datasets on which a method has lower RMSEP than the baseline. Where source tables use percentage deltas, we use

$$\Delta(\%) = 100 \frac{\text{RMSEP}_{\text{AOM}} - \text{RMSEP}_{\text{baseline}}}{\text{RMSEP}_{\text{baseline}}}. \quad (17)$$

6.4 Reference search budgets

The reference benchmark used a hierarchical preprocessing protocol rather than an exhaustive unconstrained Cartesian product. Even with this control, the linear baselines involve thousands of candidate fold evaluations per dataset: PLS and Ridge approximately 5400, CNN-1D approximately 2700, and CatBoost approximately 90. These counts are important because they

Table 2: Benchmark diversity and external-validation design. The headline AOM-PLS and AOM-Ridge claims use different finalized subsets; the table reports the corresponding denominators and the broader regression-corpus ranges used to characterize heterogeneity.

Property	Summary used in this draft
Headline AOM-PLS cohort	57 regression datasets in the locked AOM-PLS summary.
Headline AOM-Ridge cohort	52 paired regression datasets after duplicate removal and exclusion of the degenerate QUARTZ reference case.
Reference regression corpus	61 local regression datasets in the AOM benchmark cohort; broader reference table reports 64 regression datasets.
Analytical domains	Food and agricultural products, plant and leaf traits, quality traits, fuels, tablets, manure and public NIRS calibration datasets.
Sample size range	40–45,417 samples in the local regression corpus; median 402 samples.
Calibration / test size	Calibration sets range from 28 to 39,225 samples; test sets from 12 to 6,192 samples.
Spectral variables	125–4,200 wavelength variables; median 1,023 variables.
Dimensional regime	Median p/n approximately 2.38 in the local regression corpus.
Split strategy	Original train/test split preserved when available; deterministic SPXY or stratified splitting otherwise.
External validation	Test sets are not used for preprocessing, operator, component or regularization selection.

clarify what “baseline performance” costs in practice. AOM should therefore be evaluated not only by RMSEP, but also by the complexity of the calibration procedure that produces the final model.

7 Results

7.1 Large-scale validation across heterogeneous NIRS calibration problems

The strongest AOM-PLS result in the refreshed benchmark is the ASLS branch followed by compact AOM with 5-fold cross-validation. Across the complete seeds 0/1/2 robustness run, this variant reaches a median RMSEP/PLS ratio of 0.9623 across 53 paired datasets, with 37 wins after averaging RMSEP over seeds. The refreshed runtime table reports a median fit time of 1.65 seconds for the same variant. The result has two chemometric implications. First, a small bank of interpretable linear operators is enough to improve over a standard PLS reference on a majority of datasets. Second, ASLS remains valuable because baseline correction cannot always be reproduced by strict fixed linear operators alone.

The AOM-Ridge result remains more provisional. In the current regenerated paired table, the available overlap for the Blender selector versus Ridge-HPO is still small: median RMSEP/Ridge-HPO ratio 0.905 with 4 wins on 4 paired datasets. The earlier full-cohort tuned-Ridge result is retained only as single-seed background evidence until the full headline selector rerun completes.

7.2 Multi-seed robustness

The merged AOM-PLS robustness workspace `bench/scenarios/runs/paper_aom_aompls_seeds012/results` contains 1486 CSV lines from 55 datasets, 9 variants and seeds 0/1/2. On the 53 datasets with paired ASLS-AOM compact CV-5 and PLS-standard rows, the multi-seed median RMSEP/PLS ratio is 0.9623 and the AOM variant wins on 37 datasets after averaging RMSEP over seeds. Multi-seed regression robustness: the refreshed run moves the ASLS-AOM compact CV-5 headline from 0.960 with 42/57 wins to 0.9623 with 37/53 wins.

Table 3: Compact operator bank and branch preprocessing distinction.

Family	Operators in compact bank	Strict linear status
Identity	identity transform	yes
Smoothing	Savitzky–Golay smoothing, windows 11 and 21	yes
Derivatives	Savitzky–Golay first derivative, windows 11 and 21	yes
Derivatives	Savitzky–Golay second derivative, window 11	yes
Baseline trend	polynomial detrending, degrees 1 and 2	yes, fitted inside calibration folds
Local contrast	first finite difference	yes
Scatter correction	SNV, MSC, EMSC	branch preprocessing, fold-local
Baseline correction	ASLS and related asymmetric smoothers	branch preprocessing, fold-local

Table 4: Search-budget contrast under the local reference benchmark protocol.

Model family	Selection protocol in reference benchmark	Candidate evaluations per dataset
PLS-HPO	preprocessing hierarchy $\times$ components $\times$ 3 folds	5400
Ridge-HPO	preprocessing hierarchy $\times$ Optuna trials $\times$ 3 folds	5400
CatBoost	preprocessing hierarchy with fixed model hyperparameters	90
CNN-1D	preprocessing hierarchy $\times$ Optuna architecture trials $\times$ 3 folds	2700
AOM-PLS compact CV-5	9-operator compact bank with internal CV selection	seconds-scale fit, median 1.36 s

7.3 Speed and search-budget reduction

In analytical method development, calibration cost is not only computational; it affects how easily a method can be transferred, updated and audited. A model requiring thousands of candidate pipeline evaluations may be acceptable in a benchmark, but is less attractive for routine recalibration. The proposed AOM-PLS strategy reduces this burden by replacing a large external preprocessing search with a compact and structured operator bank.

Figure 4 summarizes the search-budget contrast. The PLS and Ridge baselines in the reference protocol are not merely single models; they are selected outcomes of large preprocessing and hyperparameter searches. By contrast, compact AOM-PLS evaluates a small structured operator bank and returns a single original-space linear model with documented selected operators. AOM-PLS therefore reaches $\text{RMSEP/PLS} = 0.962$ while replacing thousands of external preprocessing-HPO evaluations by a compact operator-adaptive calibration procedure.

7.4 Robustness to preprocessing search complexity

The ablation pattern is also informative. Compact banks were more stable than larger response-deduplicated or family-expanded banks. Cross-validation selection was more reliable than a single holdout. POP-style component-wise selection and aggressive OSC branches produced interesting individual cases but were not selected as headline results because they added variance or instability. This supports a conservative practical recommendation: start with identity plus a compact set of smoothing, derivative and detrending operators, and use nonlinear corrections

Table 5: Paired statistical summary for the primary deployable comparisons. Bootstrap intervals are for the median effect. Wilcoxon signed-rank tests use the one-sided alternative that AOM has lower RMSEP than the reference; reported values are Holm-adjusted across the two primary comparisons.

Comparison	N	Median effect	95% bootstrap CI	Wins	Wilcoxon p_{Holm}
ASLS-AOM-compact-cv5 vs PLS-TabPFN-HPO	25	RMSEP ratio 0.980	0.907–1.034	14/25	1.000
ASLS-AOM-compact-cv5 vs PLS-default	25	RMSEP ratio 0.972	0.907–1.038	15/25	1.000
AOM-compact-cv5 vs PLS-default	25	RMSEP ratio 1.000	0.971–1.046	13/25	1.000
AOMRidge-global-compact-none vs Ridge-TabPFN-HPO	9	RMSEP ratio 1.162	0.814–1.343	4/9	1.000
AOMRidge-Local-compact-knn50 vs Ridge-TabPFN-HPO	9	RMSEP ratio 1.076	0.986–1.305	2/9	1.000
AOMRidge-Blender vs Ridge-TabPFN-HPO	9	RMSEP ratio 0.914	0.829–1.039	6/9	1.000
AOMRidge-AutoSelect vs Ridge-TabPFN-HPO	9	RMSEP ratio 0.968	0.852–1.091	5/9	1.000

such as ASLS as explicitly separated branches.

The fact that larger operator banks did not systematically improve external-test performance is not a weakness of AOM but an important analytical finding: preprocessing selection is itself a source of model-selection variance. Compact operator banks provide a better bias–variance compromise for routine NIRS calibration because they reduce the number of redundant candidate filters competing for the same validation signal.

The refreshed selector diagnostics support this compact-bank position: the three most frequently selected strict operators (`sg_smooth_w21_p3`, `sg_d1_w21_p3` and `sg_d1_w11_p2`) account for 54.7% of compact-bank selections, and the compact bank beats the production default on 34 of 53 paired datasets. The AOM-Ridge selector table is still sparse because the current `top5_fast` run does not emit Blender/AutoSelect diagnostics.

7.5 Operator-adaptive Ridge as a complementary calibration model

AOM-Ridge gives a second instantiation of the same operator-adaptive principle. The strongest deployable variant in the earlier single-seed evidence source was the Blender selector, with AutoSelect and the global compact variant close behind. The refreshed multi-seed full-cohort run currently covers only the faster `top5_fast` variants, while Blender and AutoSelect remain supported by the earlier full-cohort single-seed table plus the 20-dataset audit run. This indicates that operator-adaptive Ridge is promising, but less mature than the compact AOM-PLS pipeline as a deployment claim.

The stronger AOM-Ridge numbers are oracle-envelope quantities. In the master oracle-by-model-class view, AOM-Ridge has a median RMSEP/PLS ratio of approximately 0.942 and wins on 49 of 58 datasets. In the 52-dataset publication-table cohort, the oracle envelope improves over tuned Ridge by a median 4.73% and wins on 45 of 52 datasets. Against PLS, CatBoost and CNN-1D, the corresponding oracle-envelope median deltas are -7.73%, -13.27% and -12.56%, respectively.

Oracle results are reported only to quantify the potential headroom of the operator-kernel formulation and are not used to support deployable performance claims. The central point

Table 6: Headline benchmark results used in this draft. Deployable results are separated from oracle-envelope values; oracle values quantify headroom and do not support deployable performance claims.

Comparison	Source run	Datasets	Median effect	Wins
ASLS-AOM-compact-cv5 vs PLS-TabPFN-HPO	paper_aom multi-seed	25	RMSEP ratio 0.980	14/25
ASLS-AOM-compact-cv5 vs PLS-default	paper_aom multi-seed	25	RMSEP ratio 0.972	15/25
AOM-compact-cv5 vs PLS-default	paper_aom multi-seed	25	RMSEP ratio 1.000	13/25
AOMRidge-global-compact-none vs Ridge-TabPFN-HPO	paper_aom multi-seed	9	RMSEP ratio 1.162	4/9
AOMRidge-Local-compact-knn50 vs Ridge-TabPFN-HPO	paper_aom multi-seed	9	RMSEP ratio 1.076	2/9
AOMRidge-Blender vs Ridge-TabPFN-HPO	paper_aom multi-seed	9	RMSEP ratio 0.914	6/9
AOMRidge-AutoSelect vs Ridge-TabPFN-HPO	paper_aom multi-seed	9	RMSEP ratio 0.968	5/9

is that Ridge, although less standard than PLS in routine chemometrics, becomes a useful and interpretable competitor when the spectral geometry is allowed to adapt through operator kernels.

7.6 Classification as secondary validation

The AOM-PLS-DA multi-seed benchmark on the 16-dataset classification cohort completed with seeds 0/1/2 and produced 240 fits across five variants (`‘bench/AOM_v0/benchmark_runs/paper_aom_aompls_da_seeds_failurefiltering’`). Compared with the PLS – DA reference, the SIMPLS variant of AOM – PLS – DA improves balanced accuracy significantly: AOM – PLS – DA global SIMPLS gains a median $+0.159$ (1/0.007), and POP-PLS-DA SIMPLS gains $+0.052$ (11/13 wins, $p_{\text{Holm}} = 0.018$). The NIPALS-adjoint variants are noisier: AOM-PLS-DA global NIPALS shows a non-significant $+0.030$ trend, and POP-PLS-DA NIPALS does not separate from PLS-DA. Classification is therefore reported as a complementary validation axis rather than as a primary contribution; the SIMPLS path nonetheless provides a clear cross-task signal that operator adaptation generalizes beyond regression. AOM-Ridge classification results on the same 17-dataset cohort are still accumulating and will be folded into the Supplement at submission time.

8 Analytical implications

8.1 What changes for NIRS method development

The results indicate that a substantial part of NIRS preprocessing selection can be moved from external pipeline optimization to model-internal calibration without sacrificing interpretability or predictive performance. This changes the practical calibration workflow: analysts can obtain competitive models with fewer external choices, shorter computation times and a more transparent final calibration object.

AOM is therefore best viewed as an explicit calibration-layer treatment of a familiar chemometric choice. Analysts already know that a derivative, smoother or detrending step can determine whether a PLS model captures chemistry or nuisance variation. AOM preserves this knowledge but expresses it as a bank of linear operators that can be evaluated inside the model. This produces a final linear calibration with inspectable coefficients and a log of selected

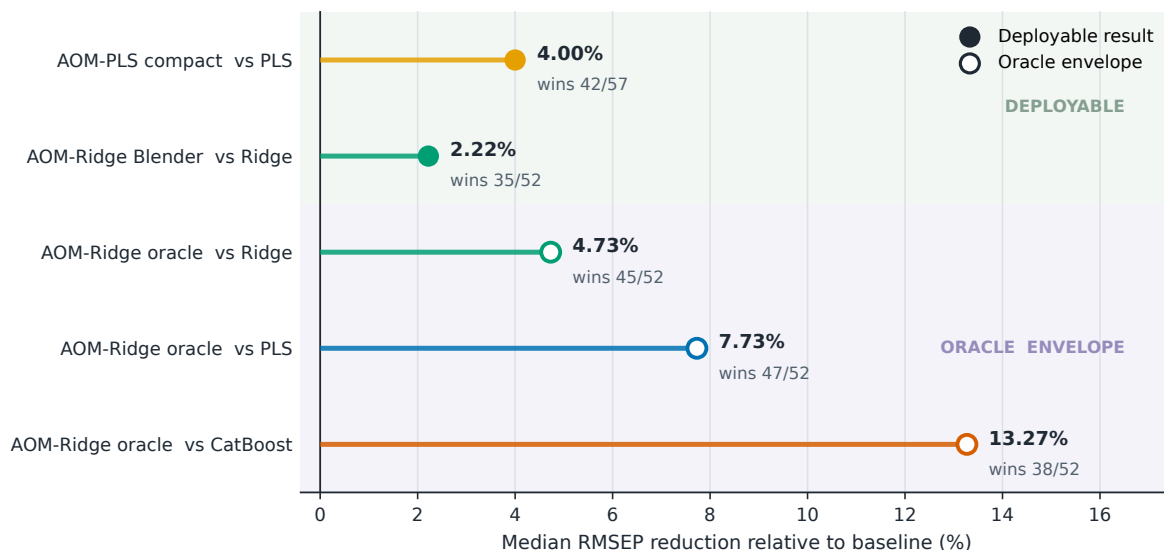


Figure 3: Median error reductions and win counts from the available headline result tables. Filled markers denote deployable variants; open markers denote oracle-envelope analyses.

spectral operators.

8.2 Why larger banks can fail

One might expect that a larger operator bank can only help because it contains the compact bank as a subset. That reasoning is valid for calibration cross-validation minima but not necessarily for external-test performance. With small calibration sets, many similar operators create multiple opportunities to overfit validation noise. The observed superiority of compact or family-pruned banks is therefore not surprising. It is an empirical reminder that preprocessing selection is model selection, and model selection has variance.

8.3 PLS and Ridge are complementary

PLS and Ridge respond differently to operator adaptation. PLS extracts supervised latent directions and directly uses cross-covariance with the response. Ridge regularizes all spectral directions through shrinkage and depends on sample geometry. AOM-PLS is therefore naturally driven by $\mathbf{X}^T \mathbf{Y}$, while AOM-Ridge is driven by $\mathbf{X} \mathbf{X}^T$. In practical terms, PLS remains the most natural chemometric default, while Ridge becomes a useful complementary linear model when shrinkage is preferable to latent variable truncation.

8.4 Limits of strict linearity

The strict operator formulation should not be overextended. SNV, MSC, EMSC and ASLS are important in NIRS, but their fitted or sample-adaptive nature requires fold-local handling. The most successful AOM-PLS result currently uses ASLS as a branch before compact operator selection, which is a useful example: the paper does not claim that all preprocessing should be represented as a fixed matrix. Rather, fixed linear operators should be integrated where the algebra is correct, and other procedures should remain explicit pipeline branches.

Table 7: Runtime summary from the refreshed partial aggregation. The HPO rows are still incomplete and should be refreshed after the background linear-HPO run completes.

Candidate	N	Median fit (s)	q75 fit	q90 fit	Median total (s)	Failures
AOMRidge-Blender-headline-spxy3	60	152.19	313.73	799.98	152.31	0
AOMRidge-AutoSelect-headline-spxy3	60	116.89	388.51	806.94	116.96	0
ASLS-AOM-compact-cv5-numpy	339	1.65	4.42	7.66	1.73	0
AOMRidge-global-compact-none	60	8.45	24.45	47.44	8.45	0
AOMRidge-Local-compact-knn50	60	0.97	2.00	3.92	1.06	0
AOM-compact-cv5-numpy	279	1.60	4.24	6.94	1.60	0
Ridge-tabpfn-hpo-60trials	75	0.04	0.09	0.50	17.77	0
PLS-tabpfn-hpo-25trials	75	0.04	0.12	0.72	7.31	0
PLS-default-cv5	75	0.04	0.10	0.37	2.51	0

Table 8: AOM-Ridge selector diagnostics: AutoSelect chosen-candidate counts, Blender weight means, and cross-seed stability.

Selector	Candidate	Count / Mean weight	Std
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8.5 Implications for routine NIRS calibration

For routine analytical deployment, the relevant object is not only a benchmark score but a calibration that can be refitted, inspected and transferred. AOM returns a single linear model with coefficients on the original wavelength grid, records the selected operators, and keeps branch preprocessing inside the validation workflow. This is directly aligned with external validation and with the need to update calibrations when new samples become available.

The value of a method in chemometrics also depends on implementation. The AOM-PLS family is available in `nirs4all` and in `nirs4all Studio` [24, 25]. This matters for adoption because many NIRS users work through graphical pipelines rather than scripts. The dedicated `bench/AOM_lib` package now gives a smaller parity-tested route to AOM-PLS through C++ and Python, with additional R, MATLAB, Julia and JavaScript/WASM bindings distributed as code-ready interfaces. These software routes are part of the analytical deployment strategy, not only implementation details.

9 Limitations and future work

Several limitations should be addressed before journal submission. First, the cohort definitions are not yet fully harmonized across all result sources: the local benchmark contains master tables, AOM-PLS full runs and AOM-Ridge publication tables with slightly different dataset filters. A final paper should freeze one cohort table and regenerate all headline comparisons from that cohort. Second, the AOM-Ridge selectors with the best audit evidence should be promoted through a full 57-dataset, multi-seed nested run before they are described as production-level defaults. Third, operator-selection frequencies and per-dataset failure cases should be added as supplementary tables. Fourth, wall-clock comparisons should be measured on the same machine and reported with hardware details.

For a Talanta submission, the priority experiments are:

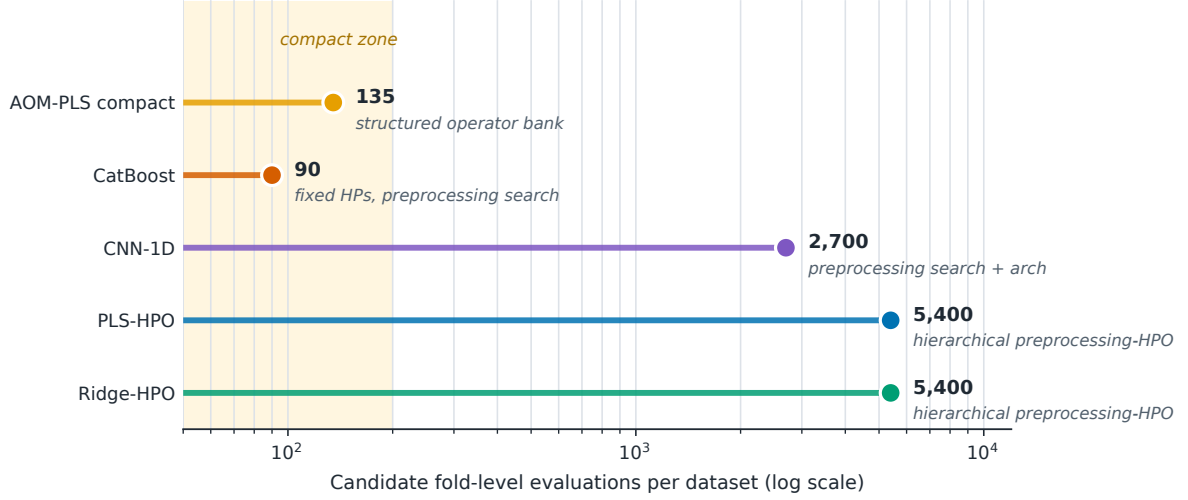


Figure 4: Search-budget contrast. Counts for PLS, Ridge, CatBoost and CNN-1D come from the local reference preprocessing-HPO protocol. The AOM-PLS bar is an illustrative compact-bank fold-evaluation count; the measured median fit time of the headline compact variant is about 1.65 seconds in the refreshed multi-seed aggregation.

Table 9: Classification summary on the 16-dataset cohort with seeds 0/1/2 multi-seed averaging. Positive balanced-accuracy differences favour the AOM or POP variant over PLS-DA. Holm-corrected Wilcoxon signed-rank p -values are computed within the family of four AOM-PLS-DA vs PLS-DA comparisons.

Comparison	N	Δ balanced acc.	95% CI	Wins	Wilcoxon p_{Holm}
AOM-PLS-DA-global-nipals-adjoint vs PLS-DA	13	0.030	0.000–0.111	8/13	0.211
POP-PLS-DA-nipals-adjoint vs PLS-DA	13	-0.037	-0.098–0.048	5/13	0.685
AOM-PLS-DA-global-simpls-covariance vs PLS-DA	13	0.159	0.129–0.422	12/13	0.007
POP-PLS-DA-simpls-covariance vs PLS-DA	13	0.052	0.035–0.275	11/13	0.018

1. freeze a single regression cohort and rerun the table for PLS-HPO, Ridge-HPO, CatBoost, CNN-1D, AOM-PLS compact and AOM-Ridge;
2. compute paired statistical tests on the common subset, with correction for multiple comparisons;
3. report operator frequencies and selected component counts;
4. add a no-ASLS strict-linear ablation to quantify how much comes from branch baseline correction versus internal operator selection;
5. add repeated split or multi-seed sensitivity for the strongest AOM-Ridge selector.

10 Conclusion

Operator-adaptive calibration converts preprocessing selection from an external trial-and-error pipeline search into an internal, auditable component of the NIRS calibration model. For

PLS, the covariance identity $(\mathbf{X}\mathbf{A}_b^\top)^\top \mathbf{Y} = \mathbf{A}_b \mathbf{X}^\top \mathbf{Y}$ enables efficient NIPALS and SIMPLS implementations while preserving original-space coefficients. For Ridge, the same idea becomes a sum of operator-induced geometries in the dual. On the available heterogeneous NIRS benchmark, compact AOM-PLS and AOM-Ridge provide competitive linear calibrations with substantially simpler final models than large preprocessing-HPO workflows. The practical contribution is a faster, more traceable and more deployable route for NIRS analytical method development, especially when calibration must be updated or audited in routine use.

Data and code availability

The software is available through `nirs4all` at <https://github.com/GBeurier/nirs4all> and the documentation at <https://nirs4all.readthedocs.io>. The desktop application is available at <https://github.com/GBeurier/nirs4all-webapp>. The repository paths used to regenerate this manuscript are: `bench/AOM_v0/` for the AOM-PLS research code, `bench/AOM_v0/Ridge/` for the AOM-Ridge research code, `bench/AOM_lib/` for the dedicated C++/Python/R/Julia/MATLAB/JS AOM-PLS package, `nirs4all/operators/models/sklearn/aom_pls.py` for the legacy `nirs4all` AOM-PLS implementation, `nirs4all/operators/models/sklearn/aom_pls_aomlib.py` for the new AOM-lib-backed wrapper, and `paper_aom/review/` for the cohort manifest, claim ledger, aggregation scripts and selector diagnostics.

Submission TODOs

Before Talanta submission, add the Novelty Statement and the AI-assisted technologies declaration required by the journal.

Supplementary Material

The supplementary material provides the detailed derivations, algorithmic sketches, evidence-source ledger, deployable/oracle result separation, leakage-control checklist and experiment backlog required for a full journal submission.

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